

Auditr Work Separation and Detailed Process Ownership

A presentation-ready explanation of who did what, how the work was divided, and which code areas each person can confidently explain.

Important use note: This is a clean and defensible division of responsibilities for viva and presentation. Adjust any wording if your actual contribution split was different.

Team summary

The project was completed jointly by Abdul Hafeedh and Maryam Hafna. The work was divided so that one person could deeply explain the backend, model, and data pipeline, while the other could deeply explain the frontend workflow, user interaction, review logic, testing flow, and presentation layer. Both contributors were involved in overall design decisions, testing, and final integration.

Suggested primary ownership

Area	Primary owner	Supporting owner	Main files	What to say if asked
App architecture and backend engine	Abdul Hafeedh	Maryam Hafna	backend/utils.py	Abdul can explain app state, auth, project persistence, CSV parsing, feature engineering, scoring, anomaly logic, and explainability helpers.
Frontend workflow and page behavior	Maryam Hafna	Abdul Hafeedh	frontend/app_shell.py and frontend/pages/*	Maryam can explain how each page serves the auditor workflow and how the UI guides the review process.
Model definition and deployment structure	Abdul Hafeedh	Maryam Hafna	backend/models/xgb_model.py	Abdul can explain that this file contains the readable Python definition of the deployed XGBoost configuration.
Training, benchmarking, and threshold selection	Abdul Hafeedh	Maryam Hafna	backend/training/train_auditr_model.py	Abdul can explain chronological holdout evaluation, model comparison, and why XGBoost was selected.
Synthetic dataset generation and demo scenario preparation	Maryam Hafna	Abdul Hafeedh	backend/training/generate_auditr_training_data.py and backend/demo/build_graduation_demo_pack.py	Maryam can explain how demo datasets were prepared for presentation and how fraud patterns were represented.
Dashboard phrasing, review flow, testing, and presentation materials	Maryam Hafna	Abdul Hafeedh	frontend/pages/Dashboard.py, Transactions.py, Explainability.py, README.md, poster and viva materials	Maryam can explain the auditor-facing flow and how the output was made understandable for presentation and review.

Area	Primary owner	Supporting owner	Main files	What to say if asked
Integration, debugging, and final validation	Both	Both	Full repo	Say that final integration, testing, and validation were shared because the full flow had to work from upload to explanation.

Detailed process ownership

Abdul Hafeedh - backend, model, and data pipeline

- Designed the backend flow in `backend/utils.py`, including project persistence, CSV parsing, feature engineering, model loading, scoring, and explanation support.
- Defined the readable deployed-model configuration in `backend/models/xgb_model.py`.
- Implemented or owned the training and evaluation pipeline in `backend/training/train_auditr_model.py`.
- Handled model benchmarking, chronological holdout evaluation, threshold selection, and saved model bundle generation.
- Worked on anomaly scoring logic using Isolation Forest with fallback normalization.
- Integrated the backend output into the frontend pages so the app could score, rank, explain, and export results correctly.

Maryam Hafna - frontend workflow, review UX, and presentation

- Focused on how the auditor experiences the workflow from project creation to review queue to case explanation.
- Explained and refined the role of pages such as Home, Projects, Dashboard, Transactions, and Explainability.
- Worked on how review outputs are framed in plain English rather than as raw model debugging data.
- Supported demo scenario preparation using the demo dataset pack and presentation-friendly audit runs.
- Prepared or owned poster, viva, documentation, and communication materials so the project could be clearly presented.
- Helped validate whether the app behavior matched the intended audit workflow, especially around uploads, queue triage, and explainability.

Shared work

- Problem selection and definition of project scope.
- Discussion of which features and fraud patterns should be represented.
- Functional testing of uploads, projects, dashboard outputs, and explainability.
- Final review of results, screenshots, poster wording, and viva preparation.

Defensible answer if a professor asks 'who did what?'

Abdul handled the main backend and machine-learning side of the project, including the feature engineering, model pipeline, scoring logic, and evaluation. Maryam handled the auditor-facing workflow, frontend explanation of the results, testing of the review flow, and presentation materials. The final integration, testing, and

overall design decisions were shared.

Which files each person should be ready to explain

- Abdul: backend/utils.py, backend/models/xgb_model.py, backend/training/train_auditr_model.py, backend/models/artifacts/model_metadata.json.
- Maryam: frontend/app_shell.py, frontend/pages/Projects.py, frontend/pages/Dashboard.py, frontend/pages/Transactions.py, frontend/pages/Explainability.py.
- Both: app.py, Launch Auditr.bat, README.md, Demo Datasets, and the overall project folder structure.